

## Lunge movement rep count error and its fix

### Before:

In the initial version of the squat logic, the app only checked if one knee reached a certain angle. However, during a lunge, the back knee often hits 90 degrees while the front knee stays higher, or vice-versa.

### Cause:

The application was unable to distinguish between a symmetrical movement (Squats) and an Asymmetrical movement (Lunge). As a result, users could cheat and perform the squat movement performing an entirely different exercise.

### Fix:

I implemented 2 layers of protection to ensure the user only performs a squat:

1. **Symmetry check (kneeDiff):** I calculated the difference between the left and right knee angles. If one knee is bent significantly more than the other (>40 degrees) then it is flagged as a lunge movement.

```
// To differentiate between a squat and a lunge
double kneeDiff = (leftKneeAngle - rightKneeAngle).abs();

// If one knee is bent significantly more than the other, then its considered a lunge
if (kneeDiff > 40 && (leftKneeAngle < 130 || rightKneeAngle < 130)) {
    feedback = "Perform a Squat, thats a Lunge!";
    if (!hasFormError) onFormError(); // Trigger voice once
    hasFormError = true;
    repHadError = true;
    errorsFound.add("Form Issues"); // Record the error type
    return;
}
```

Figure 1: Lunge Detector

2. **Bilateral Requirement:** Instead of checking if any knee is down, I used logical AND(&&) operators so both left and right knees and both hips should reach the threshold simultaneously.

```
// Bilateral Check
// uses && for both knees and both hips
if (leftKneeAngle < 100 && rightKneeAngle < 100 && leftHipAngle < 120 && rightHipAngle < 120) {
    isDown = true;
    depthAchieved = true;
    if (!hasFormError) feedback = "Now Stand Up!";
}
// check if one leg is deep but the other is not
else if ((leftKneeAngle < 100 && rightKneeAngle > 115) || (rightKneeAngle < 100 && leftKneeAngle > 115)) {
    feedback = "Balance your weight on both legs!";
    if (!hasFormError) onFormError(); // Trigger voice once
    hasFormError = true;
    repHadError = true; // if yes block the rep
    errorsFound.add("Form Issues");
}
// if the squat depth is not enough
else if (leftKneeAngle < 140 || rightKneeAngle < 140) {
    isDown = true;
    if (!depthAchieved) {
        hasFormError = true;
        feedback = "Go Lower!";
    }
}
```

Figure 2: Bilateral Requirement

If lunge is detected feedback is shown: **“Perform a Squat, that’s a lunge!”**

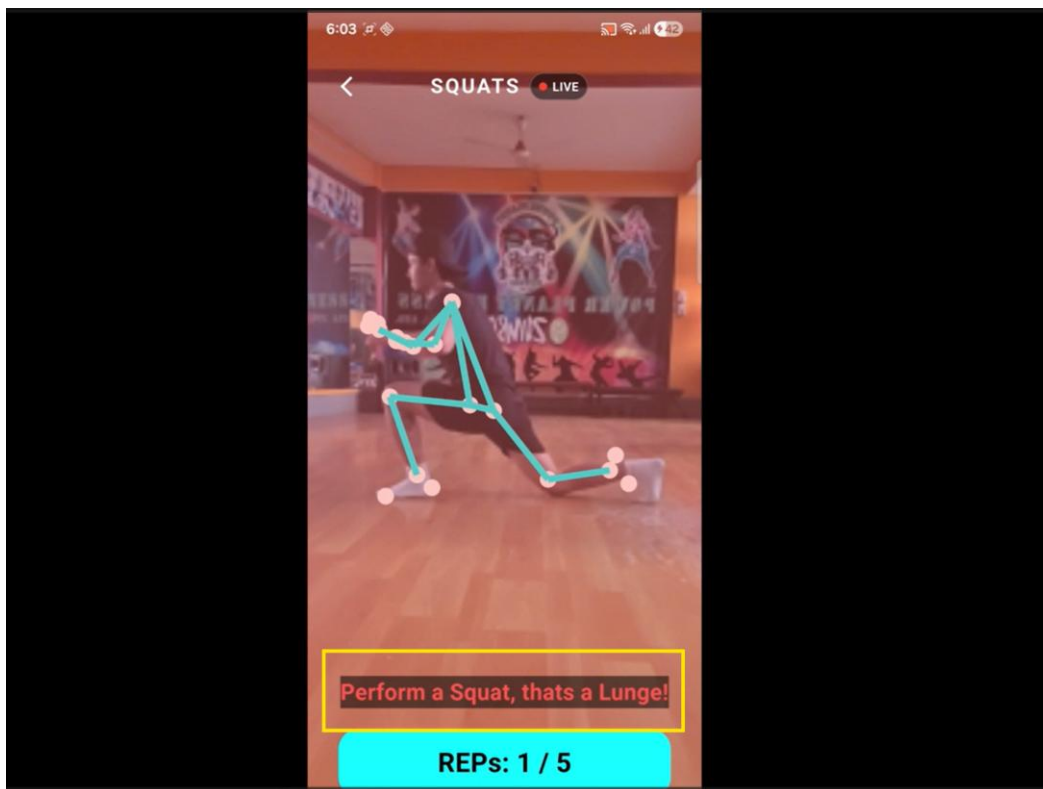
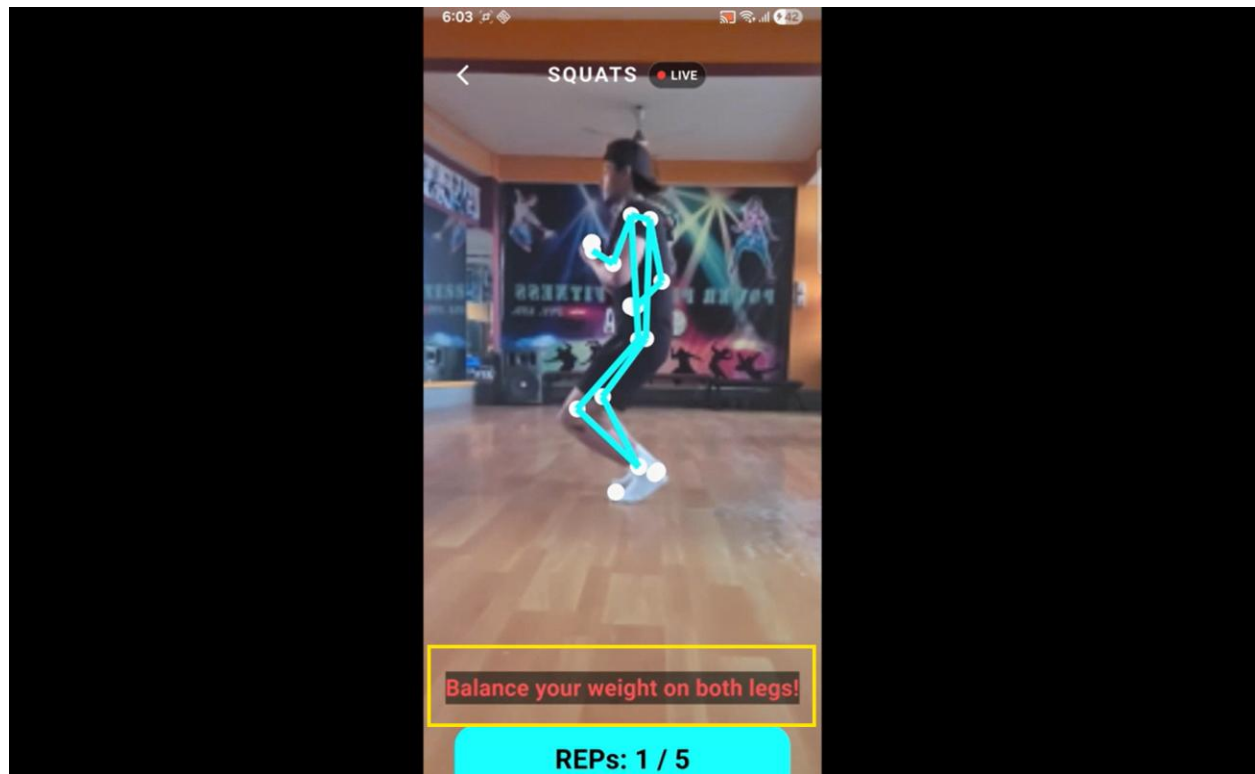


Figure 3: Lunge movement feedback

If the legs are slightly uneven feedback is shown: **“Balance your weight on both legs!”**



*Figure 4: Legs uneven feedback*